Auditing QUBO-Feasible Fuel Layouts with OpenMC: An Exploratory CB-PFR Screening Study

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***Abstract*—Composition feasibility under a binary-quadratic objective does not establish physical acceptability, and a post-physics rule cannot identify a unique candidate when the declared evidence leaves candidates tied. This paper reports an evidence-audited, exploratory software study of Confidence-Bounded Physical Feasibility Ranking (CB-PFR) on a two-dimensional multigroup C5G7-derived fuel-placement case. Four fuel positions receive exactly two mixed-oxide (MOX) and two uranium-dioxide assemblies, while five reflector positions remain fixed, producing six enumerated labels and four diagonal-reflection symmetry classes. A composition-only quadratic unconstrained binary optimization (QUBO) penalty gives every feasible label zero energy and therefore cannot rank physical responses; no pre-physics optimizer or annealer was run. An archived OpenMC campaign contains 18 runs (three seeds per label). A retrospective screen with illustrative limits $k_{\text{eff}} \in [1.14, 1.21]$, $F_{\max} = 1.35$, and width factor $z = 1$ passes two symmetry classes (three labels) and fails two classes. Since the limits were chosen after the campaign was inspected, this is an implementation demonstration, not a confirmatory test. A separate three-run high-compute UMMU screen gives mean $k_{\text{eff}} = 1.18647358$, an absolute difference of -7.64 pcm from the saved scalar reference, and a maximum single-run standard deviation of 9.48 pcm. The results support a traceable exploratory workflow but do not establish a physical optimum, calibrated uncertainty coverage, full benchmark validation, or comparative superiority.**

***Index Terms*—CB-PFR, C5G7 benchmark, OpenMC, QUBO, reproducible research software**

I. Introduction

Discrete optimization can satisfy an encoded constraint without establishing that a candidate is physically acceptable. A binary-quadratic objective distinguishes candidates only through terms it represents [1], [2]. In a fuel-layout study, a composition penalty can enforce material counts while leaving the physical responses of all feasible permutations unranked. This distinction is central to the present four-position case.

This paper studies CB-PFR (Confidence-Bounded Physical Feasibility Ranking) as a transparent post-simulation screening component in a research-software workflow. Its contribution is bounded: the implementation makes declared limits, uncertainty semantics, violations, and ties visible; the accompanying C5G7/OpenMC case study shows how those rules behave on recorded data. It is not a claim that CB-PFR is superior to established constraint-handling or stochastic ranking methods, nor that a scalable optimizer has been demonstrated. In particular, all six feasible layouts were enumerated; no pre-physics optimizer or annealer was run.

II. Case Study and Methods

A. Candidate Space and Composition QUBO

The two-dimensional model fixes five reflector positions and varies four fuel positions $p_{00}, p_{01}, p_{10}, p_{11}$. Each fuel position receives either MOX or UO_2, with exactly two of each. The six label strings are MMUU, MUMU, MUUM, UMMU, UMUM, and UUMM. Let $y_i \in \{0, 1\}$ indicate MOX at position i. The composition-only penalty is

$$E(\mathbf{y}) = A\left(\sum_{i=1}^{4} y_i - 2\right)^2, \qquad A > 0. \tag{1}$$

Using $y_i^2 = y_i$, its expansion is

$$E(\mathbf{y}) = A\left(4 - 3\sum_i y_i + 2\sum_{i<j} y_i y_j\right). \tag{2}$$

Every feasible assignment has $\sum_i y_i = 2$, hence $E = 0$. This algebraic degeneracy is a limitation of the stated composition-only objective, not of QUBO in general. The study is a complete enumeration of a small design space and does not test whether a pre-physics optimizer is useful at larger scales.

B. Symmetry and Neutronic Metrics

The declared 3×3 position map places the four variable fuel locations at the upper-left 2×2 positions and the five reflectors elsewhere. Reflection across the main diagonal swaps p_{01} and p_{10}. Thus MMUU/MUMU form class S1, MUUM is S2, UMMU is S3, and UMUM/UUMM form S4. This class reduction depends on the documented mapping and symmetric model assumptions.

OpenMC is used for multigroup neutron-transport and tally calculations [3]; the C5G7 MOX specification provides the benchmark context [4]. For the four variable fuel regions, let R_i be the assembly-integrated fission tally. The relative peaking proxy is

$$F = \frac{\max(R_1, R_2, R_3, R_4)}{\frac{1}{4}\sum_{i=1}^{4} R_i}. \tag{3}$$

F is dimensionless and relative. It is not absolute power, pin power, a thermal-hydraulic quantity, or an approved safety criterion. Source-distribution entropy and k-generation traces are used as descriptive diagnostics; monitoring source convergence is important in eigenvalue calculations, but no single scalar diagnostic alone certifies convergence [5].

C. CB-PFR Rule

For candidate or symmetry class i and metric j, let $\bar{x}_{ij}$ be the observed estimate and s_{ij} the supplied uncertainty summary. CB-PFR forms operational bounds

$$B_{ij}^{-} = \bar{x}_{ij} - z_j s_{ij}, \quad B_{ij}^{+} = \bar{x}_{ij} + z_j s_{ij}. \tag{4}$$

For declared lower and upper limits L_j, U_j, violation is

$$v_{ij} = \max(0, L_j - B_{ij}^{-}) + \max(0, B_{ij}^{+} - U_j), \tag{5}$$

including only terms for limits that are declared. The normalized violation is $V_i = \sum_j w_j v_{ij}/d_j$, where $w_j \geq 0$ and $d_j > 0$ are the metric weight and scale. These operational bounds are not calibrated confidence intervals. Missing or invalid required values fail closed.

When feasible candidates have identical substantive scores, CB-PFR reports a tie. An input-order fallback, if used for reproducible display, is not a physical preference. A separate downstream selector may apply an explicitly declared secondary score, but it does not make the composition-only pre-physics QUBO informative and is not an optimizer advantage result.

D. Archived Campaigns and Exploratory Limits

The principal layout campaign contains 18 corrected-group runs: six labels across seed rounds 820000, 830000, and 840000. Each used 7,000 total batches, 500 inactive batches, 10,000 particles per batch, and two threads. The run-level table records unique seeds and statepoint hashes and a common MGXS hash. Archive consistency checks passed; independent signed model/tally fidelity review remains unavailable.

A separate canonical UMMU high-compute screen used three independent runs (seeds 720000, 730000, and 740000), 7,000 batches, 100 inactive batches, and 10,000 particles per batch. Its saved report records the standard C5G7 energy-group structure and raw HDF5 audit checks as passing, but labels the result `NUMERICAL_SCREEN_PASS_FULL_BENCHMARK_STILL_BLOCKED`.

No design-specific acceptance rule was independently approved before the six-layout campaign. Therefore, $k_{\text{eff}} \in [1.14, 1.21]$, $F_{\max} = 1.35$, $z = 1$, scales $d_k = 0.01$, $d_F = 0.1$, and unit weights are explicitly retrospective illustrative settings. They exercise the software pathway only. A sweep over $F_{\max} \in \{1.30, 1.32, 1.35, 1.80, 1.82, 1.87, 1.90\}$ and $z \in \{0, 1, 2\}$ is sensitivity analysis, not confirmation.

III. Results

A. Symmetry-Class Aggregation

Table I summarizes class-level run results. $s_{k,\text{run}}$ is the sample standard deviation of run-level k_{eff} means within each class; $\bar{s}_{k,\text{within}}$ is the mean reported within-run standard deviation. For the illustrative CB-PFR screen, $s_k^* = \max(s_{k,\text{run}}, \bar{s}_{k,\text{within}})$. The s_F column is the sample standard deviation of run-level F observations pooled within the class. Values are descriptive, not estimates of calibrated confidence coverage.

The measured F means separate the four classes descriptively, while mirror members within S1 and S4 should not be assigned distinct physical ranks from their small sampling differences. Across the 18 layout runs, first-to-last active-quartile mean shifts ranged from -1.81×10^{-3} to $+3.11 \times 10^{-3}$ for source entropy and from -1.07×10^{-3} to $+1.16 \times 10^{-3}$ for k-generation. These summaries are diagnostic only: no convergence tolerance was set prospectively, and small quartile shifts do not certify source convergence.

B. CB-PFR Status and Threshold Sensitivity

For $z = 1$, the illustrative k_{eff} bounds all fall within $[1.14, 1.21]$, so $V_k = 0$ for all classes. Table II reports B_k^-, B_k^+, the upper operational bound B_F^+, and aggregate normalized violation V_i. Since $V_k = 0$ for all classes in this scenario, $V_i = V_F = \max(0, B_F^+ - F_{\max})/d_F$. Two classes pass (S1 and S2), represented by three layout labels; S3 and S4 fail. Since $F_{\max}$ was selected retrospectively, this result demonstrates code behavior only.

Table III reports counts as *label strings / symmetry classes*; for example, S1 contributes two label strings but one physical class. The $z = 0$ and $z = 1$ counts coincide across this tested grid; $z = 2$ changes the count only at $F_{\max} = 1.87$. This is a useful warning about threshold sensitivity, not evidence that the threshold is physically justified.

C. Separate Canonical High-Compute Numerical Screen

The independent canonical UMMU campaign used seeds 720000, 730000, and 740000 with 7,000 batches, 100 inactive batches, and 10,000 particles per batch. Its report gives mean $k_{\text{eff}} = 1.1864735755$, between-run sample SD 7.70×10^{-6}, pooled within-run reported SD 9.45×10^{-5}, and maximum individual-run SD 9.48 pcm. Against the saved scalar reference $k_{\text{ref}} = 1.18655$, the absolute difference is

$$(k_{\text{mean}} - k_{\text{ref}}) \times 10^5 = -7.64 \text{ pcm},$$

while the relative convention $(k_{\text{mean}} - k_{\text{ref}})/k_{\text{ref}} \times 10^5$ gives approximately -6.44 pcm. The conventions are distinct.

The saved project screen reports a combined-uncertainty score $z = -0.804$ using its configured reference standard uncertainty 9.5×10^{-5} in absolute k_{eff} units. This score is conditional on that stored uncertainty convention, not a published benchmark acceptance criterion. The workflow itself labels the outcome `NUMERICAL_SCREEN_PASS_FULL_BENCHMARK_STILL_BLOCKED`: independent primary-specification/model review and a benchmark-compatible spatial tally comparison remain unresolved. This scalar comparison is limited to the canonical UMMU case and does not validate the five other layout labels.

IV. Discussion

The results illustrate three distinct issues. First, the composition-only QUBO enforces the material count but ties all feasible layouts by construction. Second, CB-PFR can make operational screening decisions auditable, but a threshold

TABLE I
CORRECTED-GROUP SIX-LAYOUT CAMPAIGN AGGREGATED BY DIAGONAL-REFLECTION CLASS.

Class	Layout labels	n	Mean k_{eff}	$s_{k,\text{run}}$	$\bar{s}_{k,\text{within}}$	Mean F	s_F
S1	MMUU / MUMU	6	1.143686	7.10×10^{-5}	9.54×10^{-5}	1.295867	0.000960
S2	MUUM	3	1.156263	8.59×10^{-5}	9.53×10^{-5}	1.316149	0.001297
S3	UMMU	3	1.186499	1.40×10^{-4}	9.70×10^{-5}	1.867100	0.001541
S4	UMUM / UUMM	6	1.204665	7.94×10^{-5}	9.71×10^{-5}	1.816406	0.000673



Fig. 1. (Left) Four variable fuel positions and five fixed reflector positions in the declared abstraction; diagonal reflection exchanges p_{01} and p_{10}. (Right) Mean F by symmetry class. Class dispersion s_F is tabulated in Table I; it is not plotted because the class SDs are much smaller than the full vertical range.

TABLE II
ILLUSTRATIVE CLASS-LEVEL CB-PFR SCREEN: $k \in [1.14, 1.21]$, $F_{\max} = 1.35$, $z = 1$.

Class	B_k^-	B_k^+	B_F^+	V_i	Status
S1	1.143591	1.143781	1.296827	0.0000	Pass
S2	1.156168	1.156358	1.317446	0.0000	Pass
S3	1.186359	1.186639	1.868641	5.1864	Fail
S4	1.204568	1.204762	1.817079	4.6708	Fail

TABLE III
RETROSPECTIVE SENSITIVITY SWEEP: PASSING LABEL STRINGS / SYMMETRY CLASSES.

$F_{\max}$	$z = 0$	$z = 1$	$z = 2$
1.30	2 / 1	2 / 1	2 / 1
1.32	3 / 2	3 / 2	3 / 2
1.35	3 / 2	3 / 2	3 / 2
1.80	3 / 2	3 / 2	3 / 2
1.82	5 / 3	5 / 3	5 / 3
1.87	6 / 4	6 / 4	5 / 3
1.90	6 / 4	6 / 4	6 / 4

selected after observing the study data supports retrospective sensitivity analysis only. Third, candidate ties must be preserved at the symmetry-class level when the model and data do not distinguish mirror orientations. The secondary proxy F separates the four class means descriptively, but this small enumerated case does not demonstrate comparative benefit over a baseline or the value of running an optimizer before simulation.

The paper does not claim that the selected proxy is an engineering acceptance criterion. A stronger confirmatory study would require physical limits justified independently of this campaign, a prospectively specified analysis, independent model/tally review, and an adequate comparison against simple methods. A larger candidate space and a meaningful physics-sensitive pre-physics objective would be required to evaluate claims about optimizer utility. Constraint handling and ranking-and-selection are established research areas [6]–[8]; the contribution here is the explicit software workflow and its evidence-bounded audit, not a claim of priority for feasibility-first ranking. Other C5G7 deterministic transport studies provide methodological context but are not treated as a direct baseline for CB-PFR [9].

V. CONCLUSION

This paper reports an exploratory, evidence-audited CB-PFR workflow on six enumerated C5G7-derived fuel layouts. The composition QUBO assigns zero energy to each feasible layout; diagonal-reflection symmetry reduces six labels to four classes. Under a retrospective illustrative screen, two classes pass and two fail, while the sensitivity table makes threshold dependence explicit. A separate high-compute canonical UMMU screen yields $k_{\text{eff}} = 1.18647358$ and an absolute reference difference of -7.64 pcm, but does not establish benchmark acceptance. The evidence supports workflow traceability and transparent handling of ties, not physical optimality, calibrated confidence coverage, full validation, or method superiority.

SOFTWARE AND DATA AVAILABILITY

The accompanying anonymized review supplement contains the 18-run layout table, the canonical high-compute summary, and the scripts needed to reproduce the tabulated post-processing from those records. The project identity and permanent public release are withheld from this double-blind manuscript; no DOI or public release is claimed in this review version.

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